

VISUALIZING DATA

THREE KEY QUESTIONS

1 What **TYPE OF DATA** are you working with?



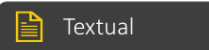
Time-series



Financial



Geospatial



Textual



Categorical



Funnel



Hierarchical



Survey



The type of data you're working with often determines **which type of visual will best represent it**; for example, using maps to represent geospatial data, line charts for time-series data, or tree maps for hierarchical data

2 What do you want to **COMMUNICATE**?

COMPARISON



Used to **compare values over time or across categories**

Common visuals:

- Column/Bar Chart
- Clustered Column/Bar
- Data Table/Heat Map
- Radar Chart
- Line Chart (*time series*)
- Area Chart (*time series*)

COMPOSITION



Used to **break down the component parts of a whole**

Common visuals:

- Stacked Bar/Column Chart
- Pie/Donut Chart
- Stacked Area (*time series*)
- Waterfall Chart (*gains/losses*)
- Funnel Chart (*stages*)
- Tree Map/sunburst (*hierarchies*)

DISTRIBUTION



Used to **show the frequency of values within a series**

Common visuals:

- Histogram
- Density Plot
- Box & Whisker
- Scatter Plot
- Data Table/Heat Map
- Map/Choropleth (*geospatial*)

RELATIONSHIP



Used to **show correlation between multiple variables**

Common visuals:

- Scatter Plot
- Bubble Chart
- Data Table/Heat Map
- Correlation Matrix

Keep it simple! While there are *hundreds* of charts to choose from, basic options like bars and columns, line charts, histograms and scatterplots often tell the simplest and clearest story

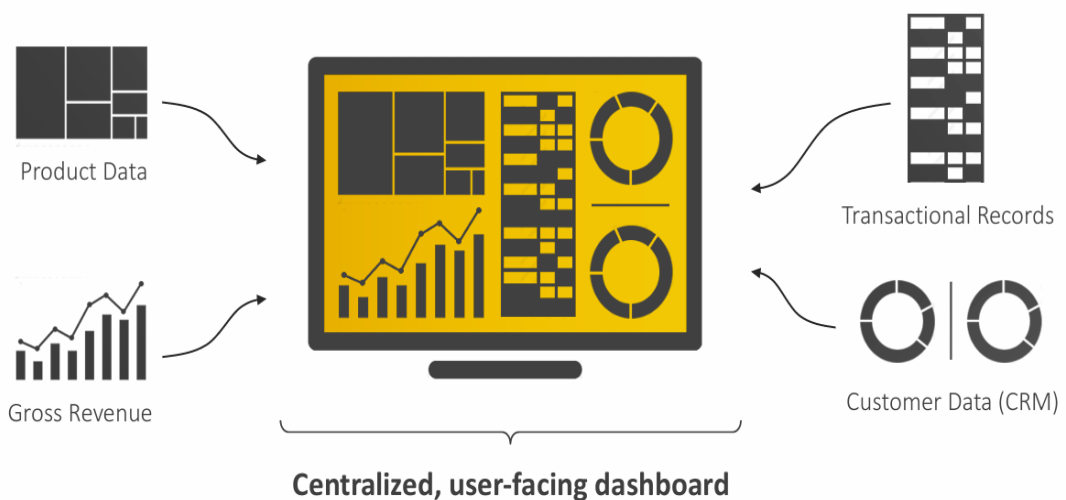
3 Who is the **END USER** and what do they need?



How you visualize and present your data is a function of **who will be consuming it**; a fellow analyst may want to see granular details, while managers and executives often prefer topline KPIs and clear, data-driven insights

ANALYTICS DASHBOARDS

Dashboards are analytics tools designed to consolidate data from multiple sources, track key metrics at a glance, and facilitate data-driven storytelling and decision making



DASHBOARD DESIGN FRAMEWORK

- 1** Define the purpose
- 2** Choose the right metrics
- 3** Present the data effectively
- 4** Eliminate clutter & noise
- 5** Use layout to focus attention
- 6** Tell a clear story

THE REPORT VIEW

Insert Menu (Add pages, visuals, buttons, shapes, images, etc.)

Report View

Report Canvas

Panes (Data, Format, Bookmarks, Selection)

Filter Pane (Page-level, report-level, visual-level filters)

Report Pages (each tab is a blank report canvas)

View Options (Zoom, fit to page)

The screenshot displays the Power BI Report View interface. The top ribbon includes tabs for File, Home, Insert, Modeling, View, Optimize, and Help. The Insert tab is active, showing options for New page, New visual, More visuals, Q&A, Key influencers, Decomposition tree, Smart narrative, Paginated report, Power Apps, Power Automate (preview), Power Platform, Text box, Buttons, Shapes, Image, and Add a sparkline. The main report canvas shows a dashboard for 'ADVENTUREWORKS' with four KPI cards: Revenue (\$24.9M), Profit (\$10.5M), Orders (25.2K), and Return Rate (2.2%). Below these are three charts: Weekly Revenue (line chart), Orders by Category (horizontal bar chart), and Top 10 Products (table). The bottom of the dashboard shows Monthly Revenue (\$1.83M), Monthly Orders (2,146), and Monthly Returns (166). The bottom navigation bar shows tabs for Exec Dashboard, Map, Product Detail, Customer Detail, Category Tooltip, AI: Q&A, AI: Decomposition Tree, and AI: Key Influencers. The right-hand pane shows the Filters pane with a search bar and sections for Filters on this page and Filters on all pages. The bottom right corner shows view options for zoom and fit to page.

Product	Orders	Revenue	Return %
Water Bottle - 30 oz.	3,983	\$39,755	1.95%
Patch 6x8 Patches	2,952	\$13,506	1.67%
Mountain Tire Tube	2,846	\$28,333	1.64%
Road Tire Tube	2,173	\$17,265	1.55%
Sport-100 Helmet, Red	2,099	\$73,444	3.33%
AWC Logo Cap	2,062	\$35,869	1.11%
Sport-100 Helmet, Blue	1,995	\$67,112	3.33%
Fender Set - Mountain	1,975	\$87,041	1.36%
Sport-100 Helmet, Black	1,940	\$65,262	2.68%
Mountain Bottle Cage	1,896	\$38,062	2.02%